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QUANTIFYING EFFECTS OF SEQUENCING VARIANCE ON CLASSIFICATION

Abstract

A method includes receiving transcriptomic data; predicting standard deviation values for each gene expression values; for each sample: computing a simulated expression value, classifying the simulated expression values, computing confidence scores; and flagging the sample; and storing the confidence scores. A computing system includes a processor; and a memory having stored thereon instructions that when executed, cause the computing system to: receive transcriptomic data; predict standard deviation values for each gene expression values; for each sample: compute a simulated expression value, classify the simulated expression values, compute confidence scores; and flag the sample; and store the confidence scores. A computer-readable media includes non-transitory computer-readable instructions that, when executed, cause a computer to: receive transcriptomic data; predict standard deviation values for each gene expression values; for each sample: compute a simulated expression value, classify the simulated expression values, compute confidence scores; and flag the sample; and store the confidence scores.

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Background/Summary

FIELD

[0001] The present disclosure is directed to methods and systems for quantifying effects of sequencing variance on classification, and more particularly, to techniques for performing role-based quality control checks on classification data.

BACKGROUND

[0002] The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

[0003] Noise and artifacts are associated with gene expression levels due to the way gene expression levels are measured. For example, RNA expression quantification techniques (e.g., a sequencing machine, a microarray or another RNA expression quantification technique) introduce variance into the RNA-seq datasets that they generate. Specifically, sequencing a sample multiple times will produce multiple datasets, wherein a single gene common to the datasets can have different expression values in each respective dataset. For example, the multiple datasets may correspond to individual datasets generated by each run of the sample. Sometimes more than one sequencing run is required to obtain multiple datasets (one run can produce multiple datasets if two “copies”/aliquots/library preparations of the sample are run together on the sequencer, wherein each “copy” corresponds to a respective generated dataset).

[0004] This noise and artifact condition may lead to indeterminate results. For example, algorithms (e.g., a machine learning model, a classifier, etc.) that use continuous measures (e.g., gene expression values measured by next-generation sequencing RNA-seq) to perform a task (e.g., a classification-based patient binning/grouping task) may produce low-confidence or indeterminate results. In the classification example, expression levels of genes used in the algorithm may be within statistical margins of error of a number of classes (e.g., subtype A, subtype B, etc.). In such a case, a sample classified as subtype A may have RNA-seq values close to a threshold between two bins (classes), and if the sample were to be sequenced a second time, the variance introduced by the sequencer may cause the RNA-seq values to be different enough that the sample could be classified as subtype B.

[0005] This discordance and indeterminateness that is the result of sequencing noise and artifacts has been empirically observed, and is a problem in multiple contexts. First, indeterminate classification has been observed during validation of patient data that required multiple sequencing runs or reactions. This caused validation to fail, and required redesign of the testing. Second, indeterminate results have been observed in a clinical setting, preventing clinicians from selecting optimal therapy regimens. Communicating discordant results to patients due to a lack of reproducible sequence-based testing has the potential to erode patient confidence.

[0006] Accordingly, there is an opportunity for improved platforms and technologies for quantifying sequence variance on classification, to quantify effects of noise and artifacts on sequencing variance.

SUMMARY

[0007] In an aspect, a computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample includes (i) receiving, via one or more processors, transcriptomic data including a plurality of observed gene

expression values each corresponding to one of a plurality of tissue samples sequenced a plurality of times; (ii) generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; (iii) for each sample in the plurality of samples: (a) computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (b) classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (c) computing, via one or more processors, a confidence score; and (d) when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample; and (iv) storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0008] In another aspect, a computing system includes one or more processors; and one or more memories, having stored thereon instructions that when executed, cause the computing system to: (i) receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of tissue samples sequenced a plurality of times; (ii) generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; (iii) for each sample in the plurality of samples: (a) compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (b) classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (c) compute, via the one or more processors, a confidence score; and (d) when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and (iv) store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0009] In yet another aspect, a computer-readable media includes a set of non-transitory computer-readable instructions that, when executed, cause a computer to: (i) receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of tissue samples sequenced a plurality of times; (ii) generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; (iii) for each sample in the plurality of samples: (a) compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (b) classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (c) compute, via the one or more processors, a confidence score; and (d) when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and (iv) store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The figures described below depict various aspects of the system and methods disclosed herein. It should be understood that each figure depicts an example of aspects of the present systems and methods.

[0011] FIG. 1 depicts an exemplary computing system for quantifying sequencer variance, according to some aspects.

[0012] FIG. 2 depicts exemplary no-call language that may be included in a report comprising indeterminate results, according to some aspects.

[0013] FIG. 3 depicts an exemplary computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample, according to some aspects.

DETAILED DESCRIPTION

Overview

[0014] The present techniques are directed to methods and systems for quantifying sampling noise during RNA expression quantification, and effects on models such as classifiers.

[0015] Empirically, “flips” between patient classes/groups/bins (for example, where more than one RNA-sequencing dataset is associated with the same sample, and the datasets are not all classified into the same group or bin) have been observed during algorithm validation. These discordant classifier results can be confusing for the clinician treating the patient, especially if each patient group is associated with a distinct therapy recommendation. In one example, the cause of these classification flips have been narrowed down to noise and artifacts in data generated during next-generation sequencing. Thus, the present techniques, which enable quantifying and predicting such flips or discordant results, provide a significant improvement to sequence data-based prediction techniques, by notifying researchers/clinicians of such potential indeterminate results, and by performing automatic mitigation.

[0016] In clinical cases (for example, where a sequenced sample is associated with a patient or organoid and the classifier result is reported to a clinician, especially to aid the clinician in selecting a therapy for a patient), it is useful to know when a sample might produce a discordant result on re-run (i.e., an indeterminate result), so that the clinician can receive accurate information to best select a therapy regimen for the patient, especially if subtype A is matched with therapy A, and subtype B is matched with therapy B. It is useful to flag these indeterminate samples when reporting results to individual patients or clinicians to ensure a high degree of patient-level reproducibility. Of course, the subtypes A and B are mere examples (any number of classes may be chosen, as in a multi-stage model).

[0017] The present techniques may use proprietary estimates of confidence in gene expression values (e.g., for the probe enhanced whole-exome capture RNA-seq platform) to predict the likelihood of a sample producing discordant subtype results upon a re-run. Herein, samples with a high likelihood of showing discordant results upon multiple re-runs are considered low-confidence or indeterminate samples. In one example, samples that fell below an 85% confidence threshold may be classified as low-confidence and reported to patients (via their clinicians) as “no-calls”. In another example, those samples may be reported as indeterminate, equivocal, etc. Conversely, samples with confidence scores greater than or equal to 85% may be classified as high confidence and thus reportable to patients/clinicians. In other embodiments, the threshold for reporting may be 60%, 65%, 70%, 75%, 80%, 90%, 95%, etc.

[0018] The present techniques may include a simulation-based approach to calculate subtype confidence scores for each sample based on its proximity to the threshold between two subtypes. In some aspects, the present simulation-based approaches may resemble those described in “Real-world data validation of the PurIST pancreatic ductal adenocarcinoma gene expression classifier and its prognostic implications,” Wenric, et al., medRxiv 2023.02.23.23286356; doi: <https://doi.org/10.1101/2023.02.23.23286356>, hereby incorporated by reference in its entirety for all purposes.

[0019] Samples with confidence scores below a threshold may be flagged as “no-calls” and not assigned a subtype; further, in some aspects, these no-call samples may be censored from analytical and clinical validations of the algorithm and may not have an associated subtype reported to

clinicians or patients.

[0020] In one aspect, the present techniques quantify confidence by using reference samples (e.g., universal human reference RNA (UHR)). The present techniques may compare noise from many (e.g., thousands or more) sequencing runs to parameterize noise specific to the sequencers used during those sequencing runs. For example, a standard deviation of UHR 20 k genes may be used to set the coefficients (this is sequencer dependent). This enables the present techniques to develop a dynamic confidence threshold for each gene. Examples of sequencers include Illumina HiSeq 4000 and Illumina NovaSeq 6000. Methods of next-generation sequencing for use in accordance with methods described herein are disclosed in Shendure 2008 Nat. Biotechnology 26:1135-1145 and Fullwood et al. 2009 Genome Res. 19:521-532, which are each hereby incorporated by reference in their respective entireties, for all purposes. Next generation sequencing methods well known in the art include synthesis technology (Illumina), pyrosequencing (454 Life Sciences), ion semiconductor technology (Ion Torrent sequencing), single-molecule real-time sequencing (Pacific Biosciences), sequencing by ligation (SOLID sequencing), nanopore sequencing (Oxford Nanopore Technologies), or paired-end sequencing. In some embodiments, massively parallel sequencing is performed using sequencing-by-synthesis with reversible dye terminators.

[0021] The present techniques are applicable to any RNA expression model that is used in a discretized fashion, and any sequencer that generates data (e.g., a set of quantified RNA expression values) that can be compared, enabling variance to be computed. Values whose variance may be quantified include gene expression levels, protein expression levels, enzyme activity, etc. However, the present techniques may be applicable to other continuous values.

[0022] In some aspects, the present techniques may be offered to another party, for example via a platform-as-a-service (PaaS) offering, software-as-a-service (SaaS), a white labeled service, etc. Such other parties may include pharmaceutical companies, biotechnical companies, contract research organizations, startups (e.g., personalized medicine firms), healthcare analytics companies, software companies, pharmacogenomics companies, healthcare consultancies, non-profit research organizations, etc.

Exemplary Computing Environments

[0023] FIG. 1 illustrates an exemplary computing environment **100** for performing the present techniques, according to some aspects. The environment **100** may include computing resources for quantifying noise and other artifacts, and for predicting whether the noise/artifacts will result in classification indeterminacy, and for further processing/computations based on such predictions.

[0024] The computing environment **100** may include a noise/artifact quantification and prediction computing device **102**, a client computing device **104**, an electronic network **106**, a sequencer system **108** and an electronic database **110**. The prediction computing device **102** may include an application programming interface **112** that enables programmatic access to the prediction computing device **102**. The components of the computing environment **100** may be communicatively connected to one another via the electronic network **106**, in some aspects. Each will now be described in greater detail.

[0025] The noise/artifact quantification and prediction computing device **102** may implement, inter alia, receipt/storage/preprocessing of quantified data (e.g., from the sequencer **108**), simulation-based algorithms, processing of quantified data, predictive tasks and related operations. In some aspects, the noise/artifact quantification and prediction computing device **102** may be implemented as one or more computing devices (e.g., one or more servers, one or more laptops, one or more mobile computing devices, one or more tablets, one or more wearable devices, one or more cloud-computing virtual instances, etc.). The noise/artifact quantification and prediction computing device **102** may include one or more processors **120**, one or more network interface controllers **122**, one or more memories **124**, an input device **126** and an output device **128**.

[0026] In some aspects, the one or more processors **120** may include one or more central processing units, one or more graphics processing units, one or more field-programmable gate

arrays, one or more application-specific integrated circuits, one or more tensor processing units, one or more digital signal processors, one or more neural processing units, one or more RISC-V processors, one or more coprocessors, one or more specialized processors/accelerators for artificial intelligence or machine learning-specific applications, one or more microcontrollers, etc.

[0027] The quantitative measures prediction computing device **102** may include one or more network interface controllers **122**, such as Ethernet network interface controllers, wireless network interface controllers, etc. The network interface controllers **122** may include advanced features, in some aspects, such as hardware acceleration, specialized networking protocols, etc.

[0028] The memories **124** of the quantitative measures prediction computing device **102** may include volatile and/or non-volatile storage media. For example, the memories **124** may include one or more random access memories, one or more read-only memories, one or more cache memories, one or more hard disk drives, one or more solid-state drives, one or more non-volatile memory express, one or more optical drives, one or more universal serial bus flash drives, one or more external hard drives, one or more network-attached storage devices, one or more cloud storage instances, one or more tape drives, etc.

[0029] The memories **124** may have stored thereon one or more modules **130**, for example, as one or more sets of computer-executable instructions. In some aspects, the modules **130** may include additional storage, such as one or more operating systems (e.g., Microsoft Windows, GNU/Linux, Mac OSX, etc.). The operating systems may be configured to run the modules **130** during operation of the noise/artifact quantification and prediction computing device **102**—for example, the modules **130** may include additional modules and/or services for receiving and processing quantitative data. The modules **130** may be implemented using any suitable computer programming language(s) (e.g., Python, JavaScript, C, C++, Rust, C#, Swift, Java, Go, LISP, Ruby, Fortran, etc.).

[0030] The modules **130** may include a quantification module **152**, a simulation module **154**, a confidence scoring module **156** and a report generation module **158**. In some aspects, more or fewer modules **130** may be included. The modules **130** may be configured to communicate with one another (e.g., via inter-process communication, via a bus, message queue, sockets, etc.).

[0031] The quantification module **152** may collect data comprising continuous data values during an experimentation process, such as during the execution of one or more sequencers operated by the sequencer system **108**. The quantification module **152** may include sets of computer-executable instructions for collecting and storing the continuous data values, for example, as a data stream. The quantification module **152** may store the collected continuous data values in the electronic database **110**. The quantification module **152** may include instructions for pre-processing the continuous data values prior to storing it, and for associating the continuous data values with other data or metadata (e.g., sample run date/time, batch number, sample identifier, etc.).

Exemplary Computer-Implemented Confidence Score Generation

[0032] The simulation module **154** may include computer-executable instructions for processing the collected continuous data values to generate confidence scores. For example, the simulation module **154** may include one or more algorithm for generating a predicted standard deviations for observed values. For example, the simulation module **154** may perform linear model computations.

[0033] In the case of the continuous data values corresponding to expression values for one or more features (e.g., one or more genes), the simulation module **154** may generate a predicted standard deviation by computing a linear model defined as

[00001] $\text{Predicted_standard_deviation} \sim a * \text{sqrt}(\log_{10}(\text{observed_expression})) - b$,

where constants a and b are determined by fitting the model to a dataset comprising a number (e.g., 1000) of replicate runs of a single sample (e.g., a universal human reference sample) on an RNA-Seq pipeline. For example, for one particular sequencing machine model, the constants a and b may be -0.08344563 and 0.2120142, respectively. The simulation module **154** may compute the standard deviation of each observed expression value for each algorithm feature (e.g., gene) using the linear model. The linear model may predict standard deviation based on mean expression alone

and may be agnostic to gene identity.

[0034] In some aspects, the one or more features may be those of PurIST, a k-top scoring pairs (k-TSP) gene signature model that assesses rank-based expression levels of 16 genes sorted into 8 gene pairs from a single sample to estimate the likelihood of that sample being basal. PurIST is described in “Purity Independent Subtyping of Tumors (PurIST), A Clinically Robust, Single-Sample Classifier for Tumor Subtyping in Pancreatic Cancer”; Clinical Cancer Research: An Official Journal of the American Association for Cancer; Research 26 (1): 82-92. Rashid et al. 2020, hereby incorporated by reference in its entirety, for all purposes. It should be appreciated that other subtyping algorithms (e.g., k-TSP based algorithms or other algorithms, single-gene models, and subtyping models including those derived from hierarchical clustering, consensus clustering, NMF, k-means, kNN and any other subtyping model making use of gene expression data to classify and label samples) may benefit from the present techniques, in addition to, or alternatively from PurIST. Generally, any model that uses any number of continuous values subject to measurement error (e.g., gene expression values) to determine discrete subtypes may be aided by the present techniques.

[0035] Other classifiers that may be used include those discussed in U.S. Pat. No. 11,527,323 entitled “Systems and methods for multi-label cancer classification,” filed on May 12, 2020 and hereby incorporated by reference herein in its entirety; and those discussed in U.S. Patent Application Publication No. 20220154284, entitled “Determination of cytotoxic gene signature and associated systems and methods for response prediction and treatment,” filed on Nov. 19, 2021, and incorporated hereby in reference in its entirety. In some aspects, the classifiers discussed in U.S. patent application Ser. No. 17/738,935, entitled “Techniques for resolving discordant her2 status in pathology images for diagnosing breast cancer” and filed on May 6, 2022, hereby incorporated by reference in its entirety, may be used. The present techniques may be extended to any subtyping algorithm, and PurIST is merely one example. For example, in some aspects, the present techniques may be used to quantify variance in gene co-expression analysis, differential expression analysis, gene network inference techniques, etc.

[0036] Next, the simulation module **154** may simulate a number of classification values, by drawing a number (e.g., 16) of gene log-transformed expression values from a number (e.g., 16) of normal distributions defined using the observed log-transformed expression values as their means along with their respective predicted standard deviations. The simulation module **154** may then apply the classifier to the simulated classifications to determine a simulated predicted classification. This process may be performed numerous times, for example, from simulations 1 N, a new set of simulated expression values may be selected from the normal distribution.

[0037] The simulation module **154** may store the simulated expression values and the simulated predicted classification in the electronic database **110**, in some aspects. The simulation module **154** may also transmit the simulated expression values and the simulated predicted classification to another element of the environment **100**, or beyond, in some aspects. Next, the simulation module **154** may repeat the simulation procedure a number (e.g., 10,000 or more) of times, and the frequencies of predicted subtypes may be counted.

[0038] The confidence scoring module **156** may include computer-executable instructions for computing one or more confidence scores. For example, the module **156** may compute one or more confidence scores by determining a ratio of each given classification to a number of total simulations. For example, if the simulation is performed by the simulation module **154** for a given sample is 10,000 times, and 9000 of the classifications classify the sample in a bin X, then the score would be 0.90 for bin X. As noted, the classifier may be a binary classifier (in which the majority of classifications may constitute the score) or a multi classifier, in which a sample is classified into two or more bins.

[0039] Generally, for a binary classifier embodiment, a score of 0.50 is a very low confidence score, being equivalent to a coin flip, while a 1.0 score is a very high confidence score, with zero

discordance expected. The confidence scoring module **156** may store the generated confidence score(s) for a given sample in the electronic database **110**, in some aspects. The confidence scoring module **156** may provide the one or more confidence scores to one or more other elements of the environment **100**, such as the report generation module **158**.

Exemplary Computer-Implemented Rules Engine

[0040] The present techniques may establish a confidence score cutoff, wherein any confidence score less than that score is considered a “no call,” meaning that the confidence level is below a predetermined threshold (e.g., 0.85). In some aspects, a no call confidence score may be used by the report generation module **158** to censor or abort communication to patients or others about corresponding classifications. The report generation module **158** may also allow “no call” classifications to be included in reports, but only with disclaimer language, “low confidence” flags, or other visual/graphical indicia, to advise the reader/viewer accordingly as to the potentially discordant status of the classification.

[0041] In some aspects, the report generation module **158** may generate static reports that include predictions regarding the confidence of classifications of patient data, as shown in FIG. 2. For example, these static reports may take the form of text documents, digital presentations, JSON documents, PDF documents, word processing documents, etc.

[0042] In some aspects, the report generation module **158** may include a rules engine for programmatically generating filtered sets of classification results based on confidence scores. For example, the electronic database **110** may include a vector or table of information corresponding to a sequencing run of a patient's tissue. In that case, the rows of the table may each include a respective gene name (e.g., as column **1**) and an associated respective expression level value (e.g., column **2**). The report generation module **158** may process that vector of information and perform its classification/prediction routines as discussed above, storing the results in the electronic database **110**.

[0043] Later, the prediction computing device **102** may receive, e.g., via the viewer application **180**, an electronic request (e.g., an HTTP POST or GET request) for classification results. The **102** may inspect the request to determine the identity of the requestor. In some aspects, the prediction computing device **102** may include a role module (not depicted) that accepts the identity of the requestor (e.g., as a session identifier, a login identifier, an authentication credential, etc.) and determines, based on that identity identifier, a role of the user (e.g., patient, clinician, research, etc.).

[0044] The rules engine of the report generation module **158** may include one or more sets of computer-executable instructions specifying which predictions the role is entitled to. The report generation module **158** may generate a query specific to the role, and issue that query against the database **110**, to retrieve only those results that the user is entitled to, based on the user's role. The report generation module **158** may then generate an electronic response (e.g., an HTTP response) to the electronic request, and transmit the filtered results to the user. In some aspects, the report generation module **158** may include “no call” classification results in this response, but only after additional rules in the rules engine annotate those results as “no call” results using a flag or other indicia, as discussed above. In some aspects, a user (e.g., a clinician) may set a preference via the viewer application **180** that affects the way that “no call” results are identified in data sets.

[0045] The present reporting techniques represent an advantageous improvement over conventional classification confidence techniques, especially because the environment **100** can be plugged into a larger computational pipeline, to provide a completely automated solution for annotation and sharing of clinical results with no-call annotations or filtering. For example, instead of the user-initiated request/response interaction discussed above, in some aspects, the application programming interface **112** may receive authenticated requests from one or more scheduled computer programs (e.g., a script that runs periodically on a remote computing device). Such script programs may be configured to have a suitable role, so that the information they receive in

responses from the application programming interface **112** is appropriately filtered/annotated.

[0046] The report generation module **158** may include computer-executable instructions for generating machine-readable results via the application programming interface **112**. For example, the application programming interface **112** may generate a structured output (e.g., results encoded in JavaScript Object Notation, XML, etc.), so that a calling client of the application programming interface **112** can easily extract data from the results, including data corresponding to the confidence score of the data.

[0047] The client computing device **104** may be accessed by a user to view results generated by the prediction computing device **102**. For example, a user may access a mobile device, laptop device, thin client, etc. embodied as the client computing device **104** to view simulation and confidence scoring results, and/or reports, with respect to a sample whose continuous values were processed by the prediction computing device **102**.

[0048] Information from the prediction computing device **102** may be transmitted via the network **106** (e.g., for display via the viewer application **180**).

[0049] The electronic network **106** may communicatively couple the elements of the environment **100**. The network **106** may include public network(s) such as the Internet, a private network such as a research institution or corporation private network, and/or any combination thereof. The network **106** may include a local area network (LAN), a wide area network (WAN), a cellular network, a satellite network, and/or other network infrastructure, whether wireless or wired.

[0050] In some aspects, the network **106** may be communicatively coupled to and/or part of a cloud-based platform (e.g., a cloud computing infrastructure). The network **106** may utilize communications protocols, including packet-based and/or datagram-based protocols such as Internet protocol, transmission control protocol, user datagram protocol, and/or other types of protocols. The network **106** may include one or more devices that facilitate network communications and/or form a hardware basis for the networks, such as one or more switches, one or more routers, one or more gateways, one or more access points (such as a wireless access point), one or more firewalls, one or more base stations, one or more repeaters, one or more backbone devices, etc.

[0051] The sequencer system **108** may include a next generation sequencer, such as a RNA sequencing whole exome capture transcriptome assay. As discussed in U.S. Pat. No. 11,043,283, entitled “Systems and methods for automating RNA expression calls in a cancer prediction pipeline,” filed Dec. 4, 2020 and in U.S. Pat. No. 11,367,508, entitled “Systems and methods for detecting cellular pathway dysregulation in cancer,” filed Aug. 14, 2020—both of which are hereby incorporated by reference in their respective entireties, for all purposes—in various embodiments, each transcriptome value set may be generated by processing a patient or tumor organoid sample through RNA whole exome next generation sequencing (NGS) to generate RNA sequencing data, and the RNA sequencing data may be processed by a bioinformatics pipeline to generate a RNA-seq expression profile for each sample. The patient sample may be a tissue sample or blood sample containing cancer cells. In more detail, RNA may be isolated from blood samples or tissue sections using commercially available reagents, for example, proteinase K, TURBO DNase-I, and/or RNA clean XP beads. The isolated RNA may be subjected to a quality control protocol to determine the concentration and/or quantity of the RNA molecules, including the use of a fluorescent dye and a fluorescence microplate reader, standard spectrofluorometer, or filter fluorometer. cDNA libraries may be prepared from the isolated RNA, purified, and selected for cDNA molecule size selection using commercially available reagents, for example Roche KAPA Hyper Beads. In another example, a New England Biolabs (NEB) kit may be used. cDNA library preparation may include the ligation of adapters onto the cDNA molecules. For example, UDI adapters, including Roche SeqCap dual end adapters, or UMI adapters (for example, full length or stubby Y adapters) may be ligated to the cDNA molecules. The sequence of nucleotides in the adapters may be specific to a sample in order to distinguish between sequencing data obtained for different samples. In this

example, adapters are nucleic acid molecules that may serve as barcodes to identify cDNA molecules according to the sample from which they were derived and/or to facilitate the next generation sequencing reaction and/or the downstream bioinformatics processing. cDNA libraries may be amplified and purified using reagents, for example, Axygen MAG PCR clean up beads. Then the concentration and/or quantity of the cDNA molecules may be quantified using a fluorescent dye and a fluorescence microplate reader, standard spectrofluorometer, or filter fluorometer. cDNA libraries may be pooled and treated with reagents to reduce off-target capture, for example Human COT-1 and/or IDT xGen Universal Blockers, before being dried in a vacufuge. Pools may then be resuspended in a hybridization mix, for example, IDT xGen Lockdown, and probes may be added to each pool, for example, IDT xGen Exome Research Panel v1.0 probes, IDT xGen Exome Research Panel v2.0 probes, other IDT probe panels, Roche probe panels, or other probes. Pools may be incubated in an incubator, PCR machine, water bath, or other temperature modulating device to allow probes to hybridize. Pools may then be processed with Streptavidin-coated beads, or another means for capturing hybridized cDNA-probe molecules, especially cDNA molecules representing exons of the human genome. In some embodiments, polyA capture may be used. Pools may be amplified and purified once more using commercially available reagents, for example, the KAPA HiFi Library Amplification kit and Axygen MAG PCR clean up beads, respectively. The cDNA library may be analyzed to determine the concentration or quantity of cDNA molecules, for example by using a fluorescent dye (for example, PicoGreen pool quantification) and a fluorescence microplate reader, standard spectrofluorometer, or filter fluorometer. The cDNA library may also be analyzed to determine the fragment size of cDNA molecules, which may be done through gel electrophoresis techniques and may include the use of a device such as a LabChip GX Touch. Pools may be cluster amplified using a kit (for example, Illumina Paired-end Cluster Kits with PhiX-spike in). In one example, the cDNA library preparation and/or whole exome capture steps may be performed with an automated system, using a liquid handling robot (for example, a SciClone NGSx). The amplification may be performed on a device, for example, an Illumina C-Bot2, and the resulting flow cell containing amplified target-captured cDNA libraries may be sequenced on a next generation sequencer, for example, an Illumina HiSeq 4000 or an Illumina NovaSeq 6000 to a unique on-target depth selected by the user, for example, 300x, 400x, 500x, 10,000x, etc. The next generation sequencer may generate a FASTQ file for each patient sample. Each FASTQ file contains reads that may be paired-end or single reads, and may be short-reads or long-reads, where each read shows one detected sequence of nucleotides in an mRNA molecule that was isolated from the patient sample, inferred by using the sequencer to detect the sequence of nucleotides contained in a cDNA molecule generated from the isolated mRNA molecules during library preparation. Each read in the FASTQ file is also associated with a quality rating. The quality rating may reflect the likelihood that an error occurred during the sequencing procedure that affected the associated read. The adapters may facilitate the binding of the cDNA molecules to anchor oligonucleotide molecules on the sequencer flow cell and may serve as a seed for the sequencing process by providing a starting point for the sequencing reaction. If two or more patient samples are processed simultaneously on the same sequencer flow cell, reads from multiple patient samples may be contained in the same FASTQ file initially and then divided into a separate FASTQ file for each patient. A difference in the sequence of the adapters used for each patient sample could serve the purpose of a barcode to facilitate associating each read with the correct patient sample and placing it in the correct FASTQ file. Each FASTQ file may be processed by a bioinformatics pipeline. In various embodiments, the bioinformatics pipeline may filter FASTQ data. Filtering FASTQ data may include correcting sequencer errors and removing (trimming) low quality sequences or bases, adapter sequences, contaminations, chimeric reads, overrepresented sequences, biases caused by library preparation, amplification, or capture, and other errors. Entire reads, individual nucleotides, or multiple nucleotides that are likely to have errors may be discarded based on the quality rating associated with the read in the FASTQ file, the

known error rate of the sequencer, and/or a comparison between each nucleotide in the read and one or more nucleotides in other reads that has been aligned to the same location in the reference genome. Filtering may be done in part or in its entirety by various software tools. FASTQ files may be analyzed for rapid assessment of quality control and reads, for example, by a sequencing data QC software such as AfterQC, Kraken, RNA-SeQC, FastQC, (see Illumina, BaseSpace Labs or <https://world.wide.web.address/illumina.com/products/by-type/informatics-products/basespace-sequence-hub/apps/fastqc.html>), or another similar software program. For paired-end reads, reads may be merged. For each FASTQ file, each read in the file may be aligned to the location in the reference genome having a sequence that best matches the sequence of nucleotides in the read. There are many software programs designed to align reads, for example, Bowtie, Burrows Wheeler Aligner (BWA), programs that use a Smith-Waterman algorithm, etc. Alignment may be directed using a reference genome (for example, GRCh38, hg38, GRCh37, other reference genomes developed by the Genome Reference Consortium, etc.) by comparing the nucleotide sequences in each read with portions of the nucleotide sequence in the reference genome to determine the portion of the reference genome sequence that is most likely to correspond to the sequence in the read. The alignment may take RNA splice sites into account. The alignment may generate a SAM file, which stores the locations of the start and end of each read in the reference genome and the coverage (number of reads) for each nucleotide in the reference genome. The SAM files may be converted to BAM files, BAM files may be sorted, and duplicate reads may be marked for deletion. In one example, kallisto software may be used for alignment and RNA read quantification (see Nicolas L Bray, Harold Pimentel, Pall Melsted and Lior Pachter, Near-optimal probabilistic RNA-seq quantification, *Nature Biotechnology* 34, 525-527 (2016), doi: 10.1038/nbt.3519). In an alternative embodiment, RNA read quantification may be conducted using another software, for example, Sailfish or Salmon (see Rob Patro, Stephen M. Mount, and Carl Kingsford (2014) Sailfish enables alignment-free isoform quantification from RNA-seq reads using lightweight algorithms. *Nature Biotechnology* (doi: 10.1038/nbt.2862) or Patro, R., Duggal, G., Love, M. I., Irizarry, R. A., & Kingsford, C. (2017). Salmon provides fast and bias-aware quantification of transcript expression. *Nature Methods*, incorporated by reference herein in its entirety for all purposes). These RNA-seq quantification methods may not require alignment. There are many software packages that may be used for normalization, quantitative analysis, and differential expression analysis of RNA-seq data. For each gene, the raw RNA read count for a given gene may be calculated. The raw read counts may be saved in a tabular file for each sample, where columns represent genes and each entry represents the raw RNA read count for that gene. In one example, kallisto alignment software calculates raw RNA read counts as a sum of the probability, for each read that the read aligns to the gene. Raw counts are therefore not integers in this example. Raw RNA read counts may then be normalized to correct for GC content and gene length, for example, using full quantile normalization and adjusted for sequencing depth, for example, using the size factor method. In one example, RNA read count normalization is conducted according to the methods disclosed in U.S. patent application Ser. No. 16/581,706 or PCT19/52801, titled Methods of Normalizing and Correcting RNA Expression Data and filed Sep. 24, 2019, incorporated by reference herein in their respective entireties. The rationale for normalization is the number of copies of each cDNA molecule in the sequencer may not reflect the distribution of mRNA molecules in the patient sample. For example, during library preparation, amplification, and capture steps, certain portions of mRNA molecules may be over or under-represented due to artifacts that arise during various aspects of priming of reverse transcription caused by random hexamers, amplification (PCR enrichment), rRNA depletion, and probe binding and errors produced during sequencing that may be due to the GC content, read length, gene length, and other characteristics of sequences in each nucleic acid molecule. Each raw RNA read count for each gene may be adjusted to eliminate or reduce over- or under-representation caused by any biases or artifacts of NGS sequencing protocols. Normalized RNA read counts may be saved in a tabular file for each sample,

where columns represent genes and each entry represents the normalized RNA read count for that gene. The electronic database **110** may include one or more suitable electronic databases for storing and retrieving data, such as relational database (e.g., MySQL databases, Oracle databases, Microsoft SQL Server databases, PostgreSQL databases, etc.). The electronic database **110** may be a NoSQL database, such as a key-value store, a graph database, a document store, etc. The electronic database **110** may be an object-oriented database, a hierarchical database, a spatial database, a time-series database, an in-memory database, etc. In some aspects, some or all of the electronic database **110** may be distributed.

[0052] In operation, one or more sequencer runs may be performed using the sequencer **108**, either by the company operating/controlling the environment **100** or by another party. The results of the sequencer are received by the prediction computing device **102**, which performs quantification **152** of the results, and optionally, storage of the quantified results. Next, the simulation module **154** processes the results, to generate confidence scores for the results. The confidence scores may be stored in association with the stored quantified results in the database **110**, for example. Next the prediction computing device **102** receives a request for data. The prediction computing device **102** provides the data, after filtering and/or annotating the data by applying rules to parameters included in the request for data. Overall, the process results in a quality control check that prevents clinicians from misinterpreting low confidence data, and prevents patients from losing confidence in the sequencing process due to the potential for passing along indeterminate results-which is advantageously largely eliminated using the present techniques, because the quality control check is performed before data is ever provided to patients.

Exemplary Computer-Implemented Report Generation

[0053] FIG. 2 depicts exemplary no-call language **200** that may be generated by the rules engine of the report generation module **158**, in some aspects. For example, the no-call language of FIG. 2 may be the language generated when the rules engine determines, via the application programming interface **112** that the requestor is a patient computing device, or that the user making the request is logged in and has a role of patient.

[0054] The various embodiments described above can be combined to provide further embodiments. All U.S. patents, U.S. patent application publications, U.S. patent application, foreign patents, foreign patent application and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified if necessary to employ concepts of the various patents, applications, and publications to provide yet further embodiments.

[0055] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Exemplary Computer-Implemented Method Aspects

[0056] FIG. 3 depicts an exemplary computer-implemented quality control method **300** for predicting whether a continuous value classifier will generate discordant classifications of a sample, according to some aspects. The method may be performed by one or more of the components of environment **100** of FIG. 1, in some aspects.

[0057] The method **300** may include receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times (block **302**). The samples may have originated from various sources, including but not limited to, tissue samples, commercially available reference RNA, manufactured RNA molecules, etc. The method **300** may include generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values (block **304**). For example, the machine learning model may be a Bayesian model,

a supervised machine learning model or an unsupervised machine learning model.

[0058] The method **300** may include for each sample in the plurality of samples (e.g., sequenced tumor samples): (i) computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; (ii) classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; (iii) computing, via one or more processors, a confidence score; and (iv) when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample (block **306**).

[0059] The method **300** may include storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data (block **308**).

[0060] In some aspects, the method **300** may include generating, via one or more processors, a static digital report including at least one classification result. In some aspects, generating the static digital report including the at least the one classification result may include censoring a no-call classification result.

[0061] In some aspects, the method **300** may include receiving, via an application programming interface, a request for sequence data (e.g., RNA sequence data); processing, via one or more processors, the request using a rules engine to determine an identity of the sender of the request; determining, via one or more processors, a role of the identity of the sender of the request; configuring, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generating, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and/or transmitting, via electronic network, the modified sequence data to a computing device of the sender of the request. In some aspects, generating the modified sequence data by processing the sequence data using the configured rules engine may generate censored results when the role of the identity of the sender of the request is a patient. In some aspects, censored results may be generated using the method **300** in other circumstances (e.g., when the results will be shared with another party that is not authorized to view uncensored results, for example due to data privacy laws). The rules engine may include additional rules governing these scenarios, in some aspects.

[0062] In some aspects, generating the modified sequence data by processing the sequence data using the configured rules engine may include generating annotated results when the role of the identity of the sender of the request is a clinician. Specifically, the annotated results may include a binary indication (e.g., call/no call) and/or additional indicia of the quality control check on the results. For example, the additional indicia may include or relate to the predicted standard deviation values, the classifications of the simulated expression values (e.g., count information), and/or the confidence scores. Whereas this information may be censored or not displayed for a patient, the physician or another viewer may have more insight into the underlying quality control techniques, and the information may prove useful for interpretation. In some aspects, the annotations may include an indication of a patient therapy, including one or more alternative therapy as discussed below.

[0063] In some aspects, the method **300** may be performed automatically as part of a bioinformatics pipeline, wherein the role of the sender of the request is a programmatic script; and wherein generating the modified sequence data by processing the sequence data may include programmatically generating a structured output corresponding to the modified sequence data, wherein the structured output includes the modified sequence data and one or more confidence scores corresponding to the modified sequence data. For example the structured output may be an XML output, a JSON output, a CSV output or another structured data format. The method **300** may include serving the structured output via an authenticated API (e.g., the API **112**) to a requesting client (e.g., the client computing device **104**) via an electronic network (e.g., the network **106**) in

some aspects.

[0064] In some aspects, the number of classifications is two or more. For example, the method **300** may include classifications with respect to two subtypes, a subtype A and a subtype B, as discussed herein. In some aspects, the classifications may be multi-stage classifications.

[0065] In some aspects, the method **300** may include determining, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be an appropriate match for the patient corresponding to the sample. The alternative therapy may include at least one of: (i) a first-line therapy or (ii) a second-line therapy. The alternative therapy may include at least one of: (i) a chemotherapeutic drug therapy, (ii) a radiation therapy, (iii) a surgery therapy, (iv) a therapy in a different class, (v) a molecular inhibitor therapy, (vi) an antibody therapy, (vii) a recombinant nucleic acid therapy, (viii) an engineered immune cell therapy, (ix) a checkpoint inhibitor therapy, (x) a cytokine therapy, (xi) a cancer treatment vaccine therapy, (xii) a CAR-cell therapy; or (xiii) an oncolytic virus therapy.

[0066] Additional exemplary therapeutic agents that can be used with the methods described herein are provided in Table 2. Additional alternative therapies are discussed in U.S. Pat. No. 11,415,571, entitled “Large scale organoid analysis,” filed on Dec. 7, 2020, and incorporated herein by reference in its entirety for all purposes.

TABLE-US-00001 TABLE 2 Agent Target Pathway Formula Veliparib (ABT-888) PARP DNA Damage C.sub.13H.sub.16N.sub.4O Selumetinib (AZD6244) MEK MAPK C.sub.17H.sub.15BrClFN.sub.4O3 PD184352 (CI-1040) MEK MAPK C.sub.17H.sub.14ClF.sub.2IN.sub.2O.sub.2 PD0325901 MEK MAPK C.sub.16H.sub.14F.sub.3IN.sub.2O.sub.4 Tozasertib (VX-680, Aurora Kinase Cell Cycle C.sub.23H.sub.28N.sub.8OS MK-0457) Y-27632 2HCl Autophagy, ROCK Cell Cycle C.sub.14H.sub.23C.sub.12N.sub.3O Olaparib (AZD2281, PARP DNA Damage C.sub.24H.sub.23FN.sub.4O.sub.3 Ku-0059436) SL-327 MEK MAPK C.sub.16H.sub.12F.sub.3N.sub.3S SB431542 TGF-beta/Smad TGF-beta/Smad C.sub.22H.sub.16N.sub.4O.sub.3 MK-2206 2HCl Akt PI3K/Akt/mTOR C.sub.25H.sub.23C.sub.12N.sub.5O Refametinib (RDEA119, MEK MAPK C.sub.19H.sub.20F.sub.3IN.sub.2O.sub.5S Bay 86-9766) KU-55933 (ATM Kinase ATM/ATR DNA Damage C.sub.21H.sub.17NO.sub.3S.sub.2 Inhibitor) GSK1904529A IGF-1R Protein Tyrosine C.sub.44H.sub.47F.sub.2N.sub.9O.sub.5S Kinase PF-04217903 c-Met Protein Tyrosine C.sub.19H.sub.16N.sub.8O Kinase U0126-EtOH MEK MAPK C.sub.20H.sub.22N.sub.6OS.sub.2 BI 2536 PLK Cell Cycle C28H39N7O3 JNJ-38877605 c-Met Protein Tyrosine C19H13F2N7 Kinase Odanacatib (MK-0822) Cysteine Protease Proteases C.sub.25H.sub.27F.sub.4N.sub.3O.sub.3S Alisertib (MLN8237) Aurora Kinase Cell Cycle C27H20ClFN4O4 Barasertib (AZD1152- Aurora Kinase Cell Cycle C26H30FN7O3 HQPA) CP-724714 EGFR, HER2 Protein Tyrosine C27H27N5O3 Kinase TGX-221 PI3K PI3K/Akt/mTOR C21H24N4O2 WZ4002 EGFR Protein Tyrosine C25H27ClN6O3 Kinase BIBR 1532 Telomerase DNA Damage C21H17NO3 Anastrozole Aromatase Endocrinology & C17H19N5 Hormones Aprepitant Substance P Others C23H21F7N4O3 TAK-700 (Orteronel) P450 (e.g. CYP17) Metabolism C18H17N3O2 PFI-1 (PF-6405761) Epigenetic Reader Epigenetics C16H17N3O4S Domain KU-0063794 mTOR PI3K/Akt/mTOR C25H31N5O4 CHIR-99021 (CT99021) GSK-3 PI3K/Akt/mTOR C22H18Cl2N8 WYE-354 mTOR PI3K/Akt/mTOR C24H29N7O5 TG100-115 PI3K PI3K/Akt/mTOR C18H14N6O2 Aurora A Inhibitor I Aurora Kinase Cell Cycle C31H31ClFN7O2 Ispinesib (SB-715992) Kinesin Cytoskeletal C30H33ClN4O2 Signaling Zibotentan (ZD4054) Endothelin Receptor GPCR & G C19H16N6O4S Protein Safinamide Mesylate MAO Metabolism C18H23FN2O5S GSK429286A ROCK Cell Cycle C21H16F4N4O2 Pimasertib (AS-703026) MEK MAPK C15H15FIN3O3 Tadalafil PDE Metabolism C22H19N3O4 Adavosertib (MK-1775) Wee1 Cell Cycle C27H32N8O2 CP-673451 PDGFR Protein Tyrosine C24H27N5O2 Kinase Selisistat (EX 527) Sirtuin Epigenetics C13H13ClN2O Dapagliflozin SGLT

GPCR & G C21H25ClO6 Protein Nebivolol HCl Adrenergic Receptor Neuronal C22H26ClF2NO4 Signaling Pimobendan PDE Metabolism C19H18N4O2 AZD8055 mTOR PI3K/Akt/mTOR C25H31N5O4 KU-60019 ATM/ATR DNA Damage C30H33N3O5S Tie2 kinase inhibitor Tie-2 ProteinTyrosine C26H21N3O2S Kinase Apixaban Factor Xa Metabolism C25H25N5O4 Raltegravir (MK-0518) Integrase Microbiology C20H21FN6O5 PCI-34051 HDAC Epigenetics C17H16N2O3 Ambrisentan Endothelin Receptor GPCR & G C22H22N2O4 Protein SB743921 HCl Kinesin Cytoskeletal C31H34Cl2N2O3 Signaling AST-1306 EGFR Protein Tyrosine C.sub.31H.sub.26ClFN.sub.4O.sub.5S Kinase Sunitinib (AZD8931) EGFR, HER2 Protein Tyrosine C23H25ClFN5O3 Kinase GSK461364 PLK Cell Cycle C27H28F3N5O2S Mubritinib (TAK 165) HER2 Protein Tyrosine C25H23F3N4O2 Kinase UK 383367 Procollagen C Metabolism C15H24N4O4 Proteinase Cryptotanshinone STAT JAK/STAT C19H20O3 Icaritin PDE Metabolism C33H40O15 OSI-027 mTOR PI3K/Akt/mTOR C21H22N6O3 Rabusertib Chk Cell Cycle C18H22BrN5O3 (LY2603618) URB597 FAAH Metabolism C20H22N2O3 A66 PI3K PI3K/Akt/mTOR C17H23N5O2S2 ICG-001 Wnt/beta-catenin Stem Cells & C33H32N4O4 Wnt PF-3845 FAAH Metabolism C24H23F3N4O2 Trametinib MEK MAPK C26H23FIN5O4 (GSK1120212) Ibrutinib (PCI-32765) BTK Angiogenesis C25H24N6O2 CHIR-124 Chk Cell Cycle C23H22ClN5O Mardepodect (PF- PDE Metabolism C25H20N4O 2545920) WAY-600 mTOR PI3K/Akt/mTOR C28H30N8O Nepicastat (SYN-117) Hydroxylase Metabolism C14H16ClF2N3S HCl RS-127445 5-HT Receptor Neuronal C17H16FN3 Signaling CP-91149 Phosphorylase Metabolism C21H22ClN3O3 SB415286 GSK-3 PI3K/Akt/mTOR C16H10ClN3O5 GSK1070916 Aurora Kinase Cell Cycle C30H33N7O Niraparib (MK-4827) PARP DNA Damage C19H20N4O CHIR-98014 GSK-3 PI3K/Akt/mTOR C20H17Cl2N9O2 AMG-458 c-Met Protein Tyrosine C30H29N5O5 Kinase Tivantinib (ARQ 197) c-Met Protein Tyrosine C23H19N3O2 Kinase Canagliflozin SGLT GPCR & G C24H25FO5S Protein NVP-BVU972 c-Met Protein Tyrosine C20H16N6 Kinase MK-5108 (VX-689) Aurora Kinase Cell Cycle C.sub.22H.sub.21ClFN.sub.3O.sub.3S SB705498 TRPV Others C17H16BrF3N4O Vistusertib (AZD2014) mTOR PI3K/Akt/mTOR C25H30N6O3 A-803467 Sodium Channel Transmembrane C19H16ClNO4 Transporters Sirtinol Sirtuin Epigenetics C26H22N2O2 Ipatasertib (GDC-0068) Akt PI3K/Akt/mTOR C24H32ClN5O2 Sapanisertib (INK 128, mTOR PI3K/Akt/mTOR C15H15N7O MLN0128) Tyrphostin AG 879 HER2 Protein Tyrosine C18H24N2OS Kinase JNJ-1661010 FAAH Metabolism C19H19N5OS CTEP (RO4956371) GluR Neuronal C19H13ClF3N3O Signaling Alogliptin (SYK-322) DPP-4 Proteases C18H21N5O2 benzoate T0070907 PPAR DNA Damage C12H8ClN3O3 GW441756 Trk receptor Protein Tyrosine C17H13N3O Kinase SB742457 5-HT Receptor Neuronal C19H19N3O2S Signaling ZM 323881 HCl VEGFR Protein Tyrosine C22H19ClFN3O2 Kinase GNF-2 Bcr-Abl Angiogenesis C18H13F3N4O2 Lumiracoxib COX Neuronal C15H13ClFNO2 Signaling JNJ-7777120 Histamine Receptor Neuronal C14H16ClN3O Signaling IOX2 HIF Angiogenesis C19H16N2O5 PF-4981517 P450 (e.g. CYP17) Metabolism C26H32N8 CHIR-99021 (CT99021) GSK-3 PI3K/Akt/mTOR C22H19Cl3N8 HCl Rivaroxaban Factor Xa Metabolism C19H18ClN3O5S Linagliptin DPP-4 Proteases C25H28N8O2 Azilsartan Medoxomil RAAS Endocrinology & C30H24N4O8 Hormones Sulfaphenazole P450 (e.g. CYP17) Metabolism C15H14N4O2S Sitagliptin phosphate DPP-4 Proteases C16H20F6N5O6P monohydrate Avanafil PDE Metabolism C23H26ClN7O3 Eprosartan Mesylate RAAS Endocrinology & C24H28N2O7S2 Hormones Carprofen COX Neuronal C15H12ClNO2 Signaling Saxagliptin hydrate DPP-4 Proteases C18H27N3O3 Daminozide Histone Demethylase Epigenetics C6H12N2O3 Bedaquiline fumarate Anti-infection Microbiology C36H35BrN2O6 JZL184 Lipase Metabolism C27H24N2O9 SC-514 IκB/IKK NF-κB C9H8N2OS2 (R)-Nepicastat HCl Hydroxylase Metabolism C14H16ClF2N3S Asunaprevir HCV Protease Proteases C35H46ClN5O9S Trelagliptin succinate DPP-4 Proteases C22H26FN5O6 Dabrafenib Mesylate Raf MAPK C.sub.24H.sub.24F.sub.3N.sub.5O.sub.5S.sub.3 Argatroban Thrombin Others C23H38N6O6S Monohydrate Sitagliptin DPP-4 Proteases C16H15F6N5O

Raltegravir potassium Integrase, HIV Microbiology C20H20FKN6O5 Protease Alogliptin DPP-4
Proteases C18H21N5O2 Dasabuvir(ABT-333) HCV Protease Proteases C26H27N3O5S
Ertugliflozin SGLT2 Ion-Channel C22H25ClO7 Rolapitant NK1-receptor GPCR C25H26F6N2O2
Dapagliflozin SGLT GPCR & G C24H35ClO9 propanediol monohydrate Protein Bedaquiline
tuberculosis Immunology C32H31BrN2O2 Fruquintinib VEGFRs VEGFR C21H19N3O5 JNJ0966
Others Others C16H16N4O2S2 acalisib (GS-9820) PI3K PI3K/Akt/mTOR C21H16FN7O BRL-
50481 PDE Metabolism C9H12N2O4S Canagliflozin SGLT GPCR & G C48H52F2O11S2
hemihydrate Protein JANEX-1 JAK JAK/STAT C16H15N3O3 Anagliptin DPP-4 Proteases
C19H25N7O2 GSK 5959 Epigenetic Reader Do Epigenetics C22H26N4O3 Pitolisant Histamine
Receptor Neuronal C17H27Cl2NO hydrochloride Signaling K 858 Kinesin Cytoskeletal
C13H15N3O2S Signaling BAY-61-3606 Syk Angiogenesis C20H20Cl2N6O3 Stattic STAT
JAK/STAT C8H5NO4S GSK2656157 PERK Apoptosis C23H21FN6O XL388 mTOR
PI3K/Akt/mTOR C23H22FN3O4S LY2090314 GSK-3 PI3K/Akt/mTOR C28H25FN6O3 MK-
8745 Aurora Kinase Cell Cycle C20H19ClFN5OS Tepotinib (EMD c-Met Protein Tyrosine
C29H28N6O2 1214063) Kinase SGC 0946 Histone Epigenetics C28H40BrN7O4
Methyltransferase GSK2334470 PDK PI3K/Akt/mTOR C25H34N8O IPA-3 PAR Cytoskeletal
C20H14O2S2 Signaling VE-822 ATM/ATR PI3K/Akt/mTOR C24H25N5O3S (+)-JQ1 Epigenetic
Reader Epigenetics C23H25ClN4O2S Domain PYR-41 E1 Activating Ubiquitin C17H13N3O7
TCID DUB Ubiquitin C9H2Cl4O2 DMH1 TGF-beta/Smad TGF-beta/Smad C24H20N4O ML347
TGF-beta/Smad, ALK TGF-beta/Smad C22H16N4O UNC1999 Histone Epigenetics C33H43N7O2
Methyltransferase SSR128129E FGFR Angiogenesis C18H15N2NaO4 Spebrutinib (CC-292, BTK
Angiogenesis C22H22FN5O3 AVL-292) SKI II S1P Receptor GPCR & G C15H11ClN2OS Protein
PF-543 S1P Receptor GPCR & G C27H31NO4S Protein CID755673 Serine/threonin Apoptosis
C12H11NO3 kinase, CaMK 1-Azakenpaullone GSK-3 PI3K/Akt/mTOR C15H10BrN3O CNX-
2006 EGFR Protein Tyrosine C26H27F4N7O2 Kinase Bisindolylmaleimide I PKC TGF-beta/Smad
C25H24N4O2 (GF109203X) Thiamet G Others Others C9H16N2O4S Alvelestat (AZD9668)
Serine Protease Proteases C24H20F3N5O4S RGFP966 HDAC Epigenetics C21H19FN4O
UNC0642 Histone Epigenetics C29H44F2N6O2 Methyltransferase NVP-TNKS656 PARP DNA
Damage C27H34N4O5 AGI-6780 Dehydrogenase Metabolism
C.sub.21H.sub.18F.sub.3N.sub.3O.sub.3S.sub.2 Ro3280 PLK Cell Cycle C27H35F2N7O3 NMS-
P937 PLK Cell Cycle C24H27F3N8O3 (NMS1286937) CNX-774 BTK Angiogenesis
C26H22FN7O3 AZD1981 GPR Endocrinology & C19H17ClN2O3S Hormones SRPIN340 Others
Others C18H18F3N3O 4µ8C Others Others C11H8O4 NMS-E973 HSP (e.g. HSP90) Cytoskeletal
C22H22N4O7 Signaling PFI-2 HCl Histone Epigenetics C23H25F4N3O3S Methyltransferase
GSK2606414 PERK Apoptosis C24H20F3N5O IPI-3063 PI3K PI3K/Akt/mTOR C25H25N7O2
Atglistatin Lipase Metabolism C17H21N3O CGP 57380 MNK MAPK C11H9FN6 SB-3CT MMP
Proteases C15H14O3S2 AR-A014418 GSK-3 PI3K/Akt/mTOR C12H12N4O4S NH125 CaMK
Neuronal C27H45IN2 Signaling XEN445 Lipase Metabolism C18H17F3N2O3R LDC000067
CDK Cell Cycle C18H18N4O3S PI-1840 Proteasome Proteases C22H26N4O3 FTI 277 HCl
Transferase Metabolism C.sub.22H.sub.30C.sub.1N.sub.3O.sub.3S.sub.2 Nexturastat A HDAC
DNA Damage C19H23N3O3 ESI-09 cAMP GPCR & G C16H15ClN4O2 Protein HJC0350 cAMP
GPCR & G C15H19NO2S Protein HO-3867 STAT JAK/STAT C28H30F2N2O2 JNK Inhibitor IX
JNK MAPK C20H16N2OS Trelagliptin DPP-4 Proteases C18H20FN5O2 XMD8-92 ERK MAPK
C26H30N6O3 A-366 Histone Epigenetics C19H27N3O2 Methyltransferase GSK-LSD1 2HCl
Histone Demethylase Epigenetics C14H22Cl2N2 LLY-507 Histone Epigenetics C36H42N6O
Methyltransferase Santacruzamate A HDAC DNA Damage C15H22N2O3 (CAY10683)
CAY10603 HDAC DNA Damage C22H30N4O6 GSK1324726A (I- Epigenetic Reader Epigenetics
C25H23ClN2O3 BET726) Domain SD-208 TGF-beta/Smad TGF-beta/Smad C17H10ClFN6
TH588 MTH1 DNA Damage C13H12Cl2N4 SB225002 CXCR GPCR & G C13H10BrN3O4
Protein CPI-360 Histone Epigenetics C25H31N3O4 Methyltransferase Picropodophyllin (PPP)

IGF-1R Protein Tyrosine C22H22O8 Kinase Savolitinib(AZD6094, c-Met Protein Tyrosine C17H15N9 HMPL-504) Kinase SP2509 Histone Demethylase Epigenetics C19H20ClN3O5S VX-11e ERK MAPK C.sub.24H.sub.20C.sub.12FN.sub.5O.sub.2 SBE 13 HCl PLK Cell Cycle C24H28Cl2N2O4 BLZ945 CSF-1R Protein Tyrosine C20H22N4O3S Kinase LFM-A13 BTK Angiogenesis C11H8Br2N2O2 EPZ015666(GSK3235025) Histone Epigenetics C20H25N5O3 Methyltransferase VER155008 HSP (e.g. HSP90) Cytoskeletal C25H23Cl2N7O4 Signaling BPTES Glutaminase Proteases C24H24N6O2S3 AZ6102 PPAR DNA Damage C25H28N6O Erlotinib EGFR Protein Tyrosine C22H23N3O4 Kinase ORY-1001 (RG-6016) Histone Demethylase Epigenetics C15H24Cl2N2 2HCl EPZ020411 2HCl Histone Epigenetics C25H40Cl2N4O3 Methyltransferase I-BRD9 Epigenetic Reader Epigenetics C.sub.22H.sub.22F.sub.3N.sub.3O.sub.3S.sub.2 Domain SirReal2 Sirtuin Epigenetics C22H20N4OS2 BDA-366 Bcl-2 Apoptosis C24H29N3O4 NVP-CGM097 Mdm2 Apoptosis C38H47ClN4O4 CC-223 mTOR PI3K/Akt/mTOR C21H27N5O3 PFI-4 Epigenetic Reader Epigenetics C21H24N4O3 Domain BIO-acetoxime GSK-3 PI3K/Akt/mTOR C18H12BrN3O3 GSK2292767 PI3K PI3K/Akt/mTOR C24H28N6O5S SIS3 HCl TGF-beta/Smad TGF-beta/Smad C28H28ClN3O3 Larotrectinib (LOXO- Trk receptor Protein Tyrosine C21H24F2N6O6S 101) sulfate Kinase PLX7904 Raf MAPK C24H22F2N6O3S VPS34-IN1 PI3K PI3K/Akt/mTOR C21H24ClN7O A-196 Histone Epigenetics C18H16Cl2N4 Methyltransferase LDC4297 (LDC044297) CDK Cell Cycle C23H28N8O SMI-4a Pim JAK/STAT C11H6F3NO2S Empagliflozin (BI SGLT GPCR & G C23H27ClO7 10773) Protein TCS 359 FLT3 Angiogenesis C18H20N2O4S NSC 23766 Rho Cell Cycle C24H38Cl3N7 GDC-0349 mTOR PI3K/Akt/mTOR C24H32N6O3 Cobimetinib (GDC- MEK MAPK C21H21F3IN3O2 0973, RG7420) GW2580 CSF-1R Protein Tyrosine C20H22N4O3 Kinase BMS-345541 IκB/IKK NF-κB C14H17N5 Dynasore Dynamin Cytoskeletal C18H14N2O4 Signaling Venetoclax (ABT-199, Bcl-2 Apoptosis C45H50ClN7O7S GDC-0199) ICI-118551 Adrenergic Receptor GPCR & G C17H28ClNO2 Hydrochloride Protein AMG 337 c-Met Protein Tyrosine C23H22FN7O3 Kinase PF-CBP1 HCl Epigenetic Reader Epigenetics C29H37ClN4O3 Domain CPI-637 Epigenetic Reader Epigenetics C22H22N6O Domain BI-78D3 JNK MAPK C13H9N5O5S2 SB366791 TRPV Transmembrane C16H14ClNO2 Transporters Thiomyristoyl Sirtuin DNA Damage C34H51N3O3S CCT245737 Chk Cell Cycle C16H16F3N7O GSK6853 Epigenetic Reader Epigenetics C22H27N5O3 Domain SHP099 phosphatase Others C16H21Cl4N5 dihydrochloride Selonsertib (GS-4997) ASK Apoptosis C24H24FN7O KYA1797K Wnt/beta-catenin Stem Cells & C17H11KN2O6S2 Wnt IPI-549 PI3K PI3K/Akt/mTOR C30H24N8O2 SGC2085 Histone Epigenetics C19H24N2O2 Methyltransferase Irbinitinib (ARRY-380, HER2 Protein Tyrosine C26H24N8O2 ONT-380) Kinase NMS-P118 PARP DNA Damage C20H24F3N3O2 BAY-876 GLUT Metabolism C24H16F4N6O2 VPS34 inhibitor 1 PI3K PI3K/Akt/mTOR C21H25N7O (Compound 19, PIK-III analogue) UK-371804 HCl Serine Protease Proteases C.sub.14H.sub.17Cl.sub.2N.sub.5O.sub.4S GSK'872 Serine/threonine kinase Apoptosis C19H17N3O2S2 (GSK2399872A) LLY-283 Histone Epigenetics C17H18N4O4 Methyltransferase GSK180736A ROCK Cell Cycle C19H16FN5O2 (GSK180736) PD-166866 (PD166866) FGFR Angiogenesis C20H24N6O3 BLU-554 (BLU554) FGFR Angiogenesis C24H24Cl2N4O4 LY3214996 ERK MAPK C22H27N7O2S PF-06651600 JAK JAK/STAT C15H19N5O FM-381 JAK JAK/STAT C24H24N6O2 AZ31 ATM/ATR DNA Damage C24H28N4O3 Tofogliflozin(CSG 452) SGLT GPCR & G C22H28O7 Protein Omarigliptin (MK-3102) DPP-4 Proteases C17H20F2N4O3S Serabelisib (INK- PI3K PI3K/Akt/mTOR C19H17N5O3 1117, MLN-1117, TAK- 117) FX1 Bcl-6 Apoptosis C14H9ClN2O4S2 Pamiparib (BGB-290) PARP DNA Damage C16H15FN4O NCT-503 Dehydrogenase Metabolism C20H23F3N4S Chk2 Inhibitor II (BML- Chk Cell Cycle C20H14ClN3O2 277) Ipragliflozin (ASP1941) SGLT GPCR & G C21H21FO5S Protein Nec-1s (7-Cl—O-Nec1) TNF-alpha Apoptosis C13H12ClN3O2 GSK'963 NF-κB, TNF-alpha NF-κB C14H18N2O GNf-6231 Wnt/beta-catenin Stem Cells & C24H25FN6O2 Wnt Skp2 inhibitor C1 CDK Cell Cycle

C.sub.18H.sub.13BrN.sub.2O.sub.4S.sub.2 (SKPin C1) PF-06840003 IDO Metabolism
 C12H9FN2O2 GI254023X Immunology & Immunology & C21H33N3O4 Inflammation related
 Inflammation BAY 1895344 (BAY- ATM/ATR DNA Damage C20H22ClN7O 1895344) Tenalisib
 (RP6530) PI3K PI3K/Akt/mTOR C23H18FN5O2 H3B-6527 FGFR Protein Tyrosine
 C29H34Cl2N8O4 Kinase Cu-CPT22 TLR Immunology & C19H22O7 Inflammation AZD1390
 ATM/ATR PI3K/Akt/mTOR C27H32FN5O2 Atuveciclib (BAY- CDK Cell Cycle C18H18FN5O2S
 1143572) LXH254 Raf MAPK C25H25F3N4O4 WM-1119 Histone Acetyltransf Epigenetics
 C18H13F2N3O3S Evobrutinib BTK Protein Tyrosine C25H27N5O2 Kinase LIT-927 CXCR
 Immunology & C17H13ClN2O3 Inflammation 4-Hydroxyquinazoline Others antiplatelet
 C8H6N2O Valbenazine tosylate VMAT2 Others C38H54N2O10S2 SAR125844 c-Met Protein
 Tyrosine C25H23FN8O2S2 Kinase dBET1 Epigenetic Reader Do Epigenetics C38H37ClN8O7S
 GSK'547 TNF-alpha Apoptosis C20H18F2N6O Palbociclib (PD- CDK Cell Cycle
 C24H30ClN7O2 0332991) HCl Palbociclib CDK Cell Cycle C26H35N7O6S (PD0332991)
 Isethionate LDN-193189 2HCl TGF-beta/Smad TGF-beta/Smad C25H23ClN6 MCC950(CP-
 456773) Immunology & Immunology & C20H23N2NaO5S Inflammation related Inflammation
 bpV (HOpic) PTEN Others C6H4K2NO8V Erlotinib HCl (OSI-744) Autophagy, EGFR Protein
 Tyrosine C22H24ClN3O4 Kinase SGX-523 c-Met Protein Tyrosine C18H13N7S Kinase (-)-
 Huperzine A (HupA) GluR, AChR Neuronal C15H18N2O Signaling GSK256066 PDE Metabolism
 C27H26N4O5S CCT137690 Aurora Kinase Cell Cycle C26H31BrN8O Capmatinib c-Met Protein
 Tyrosine C23H17FN6O (INCB28060) Kinase EPZ005687 Histone Epigenetics C32H37N5O3
 Methyltransferase GSK126 Histone Epigenetics C31H38N6O2 Methyltransferase Tazemetostat
 (EPZ- Histone Epigenetics C34H44N4O4 6438) Methyltransferase ISRIB (trans-isomer) PERK
 Apoptosis C22H24Cl2N2O4 A-1210477 Bcl-2 Apoptosis C46H55N7O7S Otenabant (CP-945598)
 Cannabinoid Receptor GPCR & G C25H26Cl3N7O HCl Protein FGF401 FGFR Protein Tyrosine
 C.sub.25H.sub.30N.sub.8O.sub.4 Kinase Lazertinib (YH25448, EGFR Protein Tyrosine
 C.sub.30H.sub.34N.sub.8O.sub.3 GNS-1480) Kinase

[0067] Aspects of the techniques described in the present disclosure may include any of the following aspects, either alone or in combination:

[0068] 1. A computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample, comprising: receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality of samples: computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; computing, via one or more processors, a confidence score; and when the confidence score is within a predetermined range, flagging the sample as a potentially discordant sample; and storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0069] 2. The computer-implemented method of aspect 1, further comprising: generating, via one or more processors, a static digital report including at least one classification result.

[0070] 3. The computer-implemented method of any of aspects 1-2, wherein generating the static digital report includes censoring a no-call classification result.

[0071] 4. The computer-implemented method of any of aspects 1-3, further comprising: receiving, via an application programming interface, a request for RNA sequence data; processing, via one or more processors, the request using a rules engine to determine an identity of a sender of the request; determining, via one or more processors, a role of the identity of the sender of the request;

configuring, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generating, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and transmitting, via electronic network, the modified sequence data to a computing device of the sender of the request.

[0072] 5. The computer-implemented method of any of aspects 1-4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating censored results when the role of the identity of the sender of the request is a patient.

[0073] 6. The computer-implemented method of any of aspects 1-4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating annotated results when the role of the identity of the sender of the request is a clinician.

[0074] 7. The computer-implemented method of any of aspects 1-4, wherein the method is performed automatically as part of a bioinformatics pipeline; wherein the role of the sender of the request is a programmatic script; and wherein generating the modified sequence data by processing the sequence data includes programmatically generating a structured output corresponding to the modified sequence data, wherein the structured output includes the modified sequence data and one or more confidence scores corresponding to the modified sequence data.

[0075] 8. The computer-implemented method of any of aspects 1-7, wherein the number of classifications is two or more.

[0076] 9. The computer-implemented method of any of aspects 1-8, further comprising: determining, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

[0077] 10. The computer-implemented method of any of aspects 1-9, wherein the alternative therapy includes at least one of: (i) a first-line therapy or (ii) a second-line therapy.

[0078] 11. The computer-implemented method of any of aspects 1-10, wherein the alternative therapy includes at least one of: (i) a chemotherapeutic drug therapy, (ii) a radiation therapy, (iii) a surgery therapy, (iv) a therapy in a different class, (v) a molecular inhibitor therapy, (vi) an antibody therapy, (vii) a recombinant nucleic acid therapy, (viii) an engineered immune cell therapy, (ix) a checkpoint inhibitor therapy, (x) a cytokine therapy, (xi) a cancer treatment vaccine therapy, (xii) a CAR-cell therapy; or (xiii) an oncolytic virus therapy.

[0079] 12. The computer-implemented method of any of aspects 1-11, wherein the machine learning model is at least one of a Bayesian model, a supervised machine learning model or an unsupervised machine learning model.

[0080] 13. A computing system, comprising: one or more processors; and [0081] one or more memories, having stored thereon instructions that when executed, cause the computing system to: receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality samples: compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; compute, via the one or more processors, a confidence score; and when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

[0082] 14. The computing system of aspect 13, the memory having stored thereon instructions that when executed, cause the computing system to: generate, via one or more processors, a static digital report including at least one classification result.

[0083] 15. The computing system of any of aspects 13-14, the memory having stored thereon

instructions that when executed, cause the computing system to: receive, via an application programming interface, a request for RNA sequence data; process, via one or more processors, the request using a rules engine to determine an identity of a sender of the request; determine, via one or more processors, a role of the identity of the sender of the request; configure, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generate, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and transmit, via electronic network, the modified sequence data to a computing device of the sender of the request.

[0084] 16. The computing system of any of aspects 13-15, the memory having stored thereon instructions that when executed, cause the computing system to: generate censored results when a role of an identity of a sender of a request is a patient.

[0085] 17. The computing system of any of aspects 13-16, the memory having stored thereon instructions that when executed, cause the computing system to: generate annotated results when a role of an identity of a sender of a request is a clinician.

[0086] 18. The computing system of aspects 13-17, wherein the number of classifications is two or more.

[0087] 19. The computing system of aspects 13-18, the memory having stored thereon instructions that when executed, cause the computing system to: determine, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

[0088] 20. A computer-readable media having stored thereon a set of non-transitory computer-readable instructions that, when executed, cause a computer to: receive, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality samples: compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classify, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; compute, via one or more processors, a confidence score; and when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

Additional Considerations

[0089] The computer-readable media may include executable computer-readable code stored thereon for programming a computer (e.g., comprising a processor(s) and GPU(s)) to the techniques herein. Examples of such computer-readable storage media include a hard disk, a CD-ROM, digital versatile disks (DVDs), an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. More generally, the processing units of the computing device **1300** may represent a CPU-type processing unit, a GPU-type processing unit, a TPU-type processing unit, a field-programmable gate array (FPGA), another class of digital signal processor (DSP), or other hardware logic components that can be driven by a CPU.

[0090] A system for performing the methods described herein may include a computing device, and more particularly may be implemented on one or more processing units, for example, Central Processing Units (CPUs), and/or on one or more or Graphical Processing Units (GPUs), including clusters of CPUs and/or GPUs. Features and functions described may be stored on and implemented from one or more non-transitory computer-readable media of the computing device.

The computer-readable media may include, for example, an operating system and software modules, or “engines,” that implement the methods described herein. Those engines may be stored as sets of non-transitory computer-executable instructions. The computing device may be a distributed computing system, such as an Amazon Web Services, Google Cloud Platform Microsoft Azure, or other public, private and/or hybrid cloud computing solution.

[0091] The computing device includes a network interface communicatively coupled to network, for communicating to and/or from a portable personal computer, smart phone, electronic document, tablet, and/or desktop personal computer, or other computing devices. The computing device further includes an I/O interface connected to devices, such as digital displays, user input devices, etc.

[0092] The functions of the engines may be implemented across distributed computing devices, etc. connected to one another through a communication link. In other examples, functionality of the system may be distributed across any number of devices, including the portable personal computer, smart phone, electronic document, tablet, and desktop personal computer devices shown. The computing device may be communicatively coupled to the network and another network. The networks may be public networks such as the Internet, a private network such as that of a research institution or a corporation, or any combination thereof. Networks can include, local area network (LAN), wide area network (WAN), cellular, satellite, or other network infrastructure, whether wireless or wired. The networks can utilize communications protocols, including packet-based and/or datagram-based protocols such as Internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), or other types of protocols. Moreover, the networks can include a number of devices that facilitate network communications and/or form a hardware basis for the networks, such as switches, routers, gateways, access points (such as a wireless access point as shown), firewalls, base stations, repeaters, backbone devices, etc.

[0093] The computer-readable media may include executable computer-readable code stored thereon for programming a computer (for example, comprising a processor(s) and GPU(s)) to the techniques herein. Examples of such computer-readable storage media include a hard disk, a CD-ROM, digital versatile disks (DVDs), an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. More generally, the processing units of the computing device may represent a CPU-type processing unit, a GPU-type processing unit, a field-programmable gate array (FPGA), another class of digital signal processor (DSP), or other hardware logic components that can be driven by a CPU.

[0094] Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components or multiple components.

[0095] Additionally, certain aspects are described herein as including logic or a number of routines, subroutines, applications, or instructions. These may constitute either software (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware. In hardware, the routines, etc., are tangible units capable of performing certain operations and may be configured or arranged in a certain manner. In example aspects, one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations

as described herein.

[0096] In various aspects, a hardware module may be implemented mechanically or electronically. For example, a hardware module may comprise dedicated circuitry or logic that is permanently configured (e.g., as a special-purpose processor, such as a microcontroller, field programmable gate array (FPGA) or an application-specific integrated circuit (ASIC)) to perform certain operations. A hardware module may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

[0097] Accordingly, the term “hardware module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering aspects in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where the hardware modules comprise a general-purpose processor configured using software, the general-purpose processor may be configured as respective different hardware modules at different times. Software may accordingly configure a processor, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

[0098] Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple of such hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connects the hardware modules. In aspects in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information).

[0099] The various operations of the example methods described herein can be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example aspects, comprise processor-implemented modules.

[0100] Similarly, the methods or routines described herein may be at least partially processor-implemented. For example, at least some of the operations of a method can be performed by one or more processors or processor-implemented hardware modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but also deployed across a number of machines. In some example aspects, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other aspects the processors may be distributed across a number of locations.

[0101] The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but also deployed across a number of

machines. In some example aspects, the one or more processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example aspects, the one or more processors or processor-implemented modules may be distributed across a number of geographic locations.

[0102] Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information.

[0103] As used herein any reference to “one aspect” or “an aspect” means that a particular element, feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. The appearances of the phrase “in one aspect” in various places in the specification are not necessarily all referring to the same aspect.

[0104] Some aspects may be described using the expression “coupled” and “connected” along with their derivatives. For example, some aspects may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. The aspects are not limited in this context.

[0105] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0106] In addition, use of the “a” or “an” are employed to describe elements and components of the aspects herein. This is done merely for convenience and to give a general sense of the description. This description, and the claims that follow, should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0107] This detailed description is to be construed as an example only and does not describe every possible aspect, as describing every possible aspect would be impractical, if not impossible. One could implement numerous alternate aspects, using either current technology or technology developed after the filing date of this application.

Claims

1. A computer-implemented quality control method for predicting whether a continuous value classifier will generate discordant classifications of a sample, comprising: receiving, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generating, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality of samples: computing, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classifying, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; computing, via one or more processors, a confidence score; and when the confidence score is within a predetermined range, flagging the

sample as a potentially discordant sample; and storing, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

2. The computer-implemented method of claim 1, further comprising: generating, via one or more processors, a static digital report including at least one classification result.

3. The computer-implemented method of claim 2, wherein generating the static digital report includes censoring a no-call classification result.

4. The computer-implemented method of claim 1, further comprising: receiving, via an application programming interface, a request for RNA sequence data; processing, via one or more processors, the request using a rules engine to determine an identity of a sender of the request; determining, via one or more processors, a role of the identity of the sender of the request; configuring, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generating, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and transmitting, via electronic network, the modified sequence data to a computing device of the sender of the request.

5. The computer-implemented method of claim 4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating censored results when the role of the identity of the sender of the request is a patient.

6. The computer-implemented method of claim 4, wherein generating the modified sequence data by processing the sequence data using the configured rules engine includes generating annotated results when the role of the identity of the sender of the request is a clinician.

7. The computer-implemented method of claim 4, wherein the method is performed automatically as part of a bioinformatics pipeline; wherein the role of the sender of the request is a programmatic script; and wherein generating the modified sequence data by processing the sequence data includes programmatically generating a structured output corresponding to the modified sequence data, wherein the structured output includes the modified sequence data and one or more confidence scores corresponding to the modified sequence data.

8. The computer-implemented method of claim 1, wherein the number of classifications is two or more.

9. The computer-implemented method of claim 1, further comprising: determining, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

10. The computer-implemented method of claim 9, wherein the alternative therapy includes at least one of: (i) a first-line therapy or (ii) a second-line therapy.

11. The computer-implemented method of claim 9, wherein the alternative therapy includes at least one of: (i) a chemotherapeutic drug therapy, (ii) a radiation therapy, (iii) a surgery therapy, (iv) a therapy in a different class, (v) a molecular inhibitor therapy, (vi) an antibody therapy, (vii) a recombinant nucleic acid therapy, (viii) an engineered immune cell therapy, (ix) a checkpoint inhibitor therapy, (x) a cytokine therapy, (xi) a cancer treatment vaccine therapy, (xii) a CAR-cell therapy; or (xiii) an oncolytic virus therapy.

12. The computer-implemented method of claim 1, wherein the machine learning model is at least one of a Bayesian model, a supervised machine learning model or an unsupervised machine learning model.

13. A computing system, comprising: one or more processors; and one or more memories, having stored thereon instructions that when executed, cause the computing system to: receive, via the one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality samples: compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene

and the predicted standard deviation value for the gene; classify, via the one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; compute, via the one or more processors, a confidence score; and when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.

14. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to: generate, via one or more processors, a static digital report including at least one classification result.

15. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to: receive, via an application programming interface, a request for RNA sequence data; process, via one or more processors, the request using a rules engine to determine an identity of a sender of the request; determine, via one or more processors, a role of the identity of the sender of the request; configure, via one or more processors, a rules engine based on the role of the identity of the sender of the request; generate, via one or more processors, modified sequence data by processing the sequence data using the configured rules engine; and transmit, via electronic network, the modified sequence data to a computing device of the sender of the request.

16. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to: generate censored results when a role of an identity of a sender of a request is a patient.

17. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to: generate annotated results when a role of an identity of a sender of a request is a clinician.

18. The computing system of claim 13, wherein the number of classifications is two or more.

19. The computing system of claim 13, the memory having stored thereon instructions that when executed, cause the computing system to: determine, based on flagging the sample as the potentially discordant sample, an alternative therapy that may be appropriate for a patient corresponding to the sample.

20. A computer-readable media having stored thereon a set of non-transitory computer-readable instructions that, when executed, cause a computer to: receive, via one or more processors, transcriptomic data including a plurality of observed gene expression values each corresponding to one of a plurality of samples sequenced a plurality of times; generate, via a machine learning model, a predicted standard deviation value for each of the plurality of observed gene expression values; for each sample in the plurality samples: compute, for each simulation in a plurality of simulations, and for each of a plurality of genes, a respective simulated expression value, by drawing from a normal distribution based on an observed expression value for the gene and the predicted standard deviation value for the gene; classify, via one or more processors, the simulated expression values using the classifier, wherein classifying the simulated expression values includes counting a number of classifications; compute, via one or more processors, a confidence score; and when the confidence score is within a predetermined range, flag the sample as a potentially discordant sample; and store, in an electronic database, at least one of the confidence scores in association with the transcriptomic data.
